

Analytic state-dependent-delay data can produce a smooth nonanalytic persisted periodic orbit

Candidate full counterexample to Remark 6.5 of arXiv:2103.05203

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Status: candidate full negative answer, with exact-rational certificate. We give an analytic three-dimensional state-dependent-delay equation, arbitrarily close to an analytic ODE with a nondegenerate limit cycle, whose unique nearby periodic orbit is C^∞ but not real analytic. The only computer-assisted step is finite exact rational arithmetic; all infinite tails have symbolic majorants recorded below and checked by the included script.

Source question

Yang, Gimeno, and de la Llave prove persistence and smooth parameter dependence of periodic orbits for functional differential equations close to an ODE. After observing that analytic state-dependent-delay data bootstrap the orbit to C^∞ , Remark 6.5 asks whether the parameterization K must be analytic and conjectures that it may fail to be so [1]. The remark is reproduced from PDF page 18.

singular and seems to require extra assumptions and slightly different proofs. The extension of the results to the small delay case is done in Section 8

Remark 6.5. *In the case of state dependent delay or distributed delay with a smooth f and r , it is automatic to show that if K is C^ℓ , the right hand side of (4) is C^ℓ , hence, looking at the left hand side of (4), K is $C^{\ell+1}$. The bootstrap stops only when we do not have any more regularity of f or r .*

So, in case that f and r are C^∞ , we obtain that the K is C^∞ .

One natural question that deserves more study is whether in the case that f and r are analytic, the K is analytic. The remarkable paper [MPN14] contains obstructions that show that equations with time dependent delays – heuristically better behaved than the ones considered here, may fail to have analytic solutions. In view of these results, it is natural to conjecture that the periodic solutions produced here, could fail to be analytic even if f and r are analytic.

The counterexample

For $X = (u, v, z) \in \mathbb{R}^3$, take

$$\dot{u} = -v + (1 - u^2 - v^2)u, \tag{1}$$

$$\dot{v} = u + (1 - u^2 - v^2)v, \tag{2}$$

$$\dot{z} = -z + v + \varepsilon z(t - r(X(t))), \quad r(u, v, z) = 2\pi - \arctan v. \tag{3}$$

This has precisely the source's form

$$\dot{X}(t) = f(X(t)) + \varepsilon P(X(t), X(t - r(X(t)))), \quad P(X, Y) = (0, 0, Y_3).$$

The maps f, P, r are real analytic, and the delay is globally bounded and strictly positive: $3\pi/2 < r < 5\pi/2$.

At $\varepsilon = 0$ there is the analytic 2π -periodic orbit

$$K_0(t) = \left(\cos t, \sin t, \frac{\sin t - \cos t}{2} \right). \quad (4)$$

Its Floquet multipliers are

$$1, \quad e^{-4\pi}, \quad e^{-2\pi}.$$

Indeed, the first two equations have radial equation $\dot{\rho} = \rho(1 - \rho^2)$ and angular equation $\dot{t} = 1$; the z -variation is triangular with diagonal exponent -1 . Thus the unit multiplier is simple and the orbit is nondegenerate in the source's sense.

The clock equations are unaffected by the perturbation. A nearby orbit with the phase fixed by $(u, v) = (\cos t, \sin t)$ therefore has third component $y = y_\varepsilon$ satisfying

$$y'(t) + y(t) = \sin t + \varepsilon y(\eta(t)), \quad \eta(t) = t + \arctan(\sin t), \quad (5)$$

where periodicity was used to replace $t - 2\pi + \arctan(\sin t)$ by $\eta(t)$.

Lemma 1 (Existence, uniqueness, and smoothness). *For every $|\varepsilon| < 1$, equation (5) has a unique continuous 2π -periodic solution. It is C^∞ and*

$$\|y_\varepsilon\|_\infty \leq \frac{1}{1 - |\varepsilon|}. \quad (6)$$

Moreover $y_\varepsilon \rightarrow y_0 = (\sin t - \cos t)/2$ as $\varepsilon \rightarrow 0$.

Proof. Let $A = D + 1$ on periodic functions and let $Cy = y \circ \eta$. The unique periodic solution of $Ay = g$ is

$$(A^{-1}g)(t) = \frac{1}{1 - e^{-2\pi}} \int_{t-2\pi}^t e^{-(t-s)} g(s) ds,$$

so $\|A^{-1}g\|_\infty \leq \|g\|_\infty$ (also for complex-valued functions), while $\|C\| = 1$. Hence

$$y = A^{-1} \sin + \varepsilon A^{-1} Cy$$

is a contraction for $|\varepsilon| < 1$, and (6) follows. The differential equation then bootstraps a continuous solution successively to every finite differentiability order. The fixed-point equation also gives convergence to $A^{-1} \sin = y_0$ as $\varepsilon \rightarrow 0$. $\square$

It remains to prove that suitable arbitrarily small $\varepsilon > 0$ give a nonanalytic solution. The expansive fixed point $\eta(0) = 0$, $\eta'(0) = 2$ supplies the obstruction.

A certified Taylor obstruction

Write the formal Taylor series at zero as

$$y(t) = \sum_{n \geq 0} y_n t^n, \quad e_{k,n} = [t^n] \eta(t)^k,$$

and, for $\varepsilon \neq 0$, normalize

$$w_n = \frac{n! y_n}{\varepsilon^n 2^{n(n-1)/2}}. \quad (7)$$

Since $\eta(t) = 2t + O(t^3)$, coefficient comparison in (5) gives

$$w_{n+1} - w_n = -\frac{w_n}{\varepsilon 2^n} + \sum_{k=0}^{n-1} A_{n,k} w_k + G_n, \quad (8)$$

$$A_{n,k} = e_{k,n} \frac{n!}{k!} \varepsilon^{k-n} 2^{k(k-1)/2 - n(n+1)/2}, \quad (9)$$

$$G_n = \frac{n! [t^n] \sin t}{\varepsilon^{n+1} 2^{n(n+1)/2}}. \quad (10)$$

Only even differences $d = n - k \geq 2$ occur in the sum because η is odd.

We record a fully explicit convergence estimate specialized to $\varepsilon = 1/10$. Put $\eta(t) = 2t(1 + u(t))$. On $|t| = 1/2$,

$$|u(t)| \leq \frac{\operatorname{arctanh}(\sinh(1/2)) - 1/2}{2(1/2)} < \frac{1}{10}. \quad (11)$$

The first inequality follows by majorizing separately the power series of $\arctan z - z$ and $\sin t - t$; the second follows, for example, from $\sinh(1/2) < 8/15$ and $\operatorname{arctanh}(8/15) = \frac{1}{2} \log(23/7) < 3/5$. Thus Cauchy's estimate and $n!/k! \leq n^{n-k}$ give

$$|A_{n,n-d}| \leq T_{n,d} := \left(\frac{11}{10}\right)^{n-d} (20n)^d 2^{-d(2n-d+1)/2}. \quad (12)$$

As a function of even d , $\log T_{n,d}$ is convex. For $n \geq 30$ its maximum is therefore at $d = 2$ or at the last admissible endpoint. Exact comparison at $n = 30, 31$, followed by

$$\frac{(T_{n+2,d_{\max}}/T_{n+2,2})}{(T_{n,d_{\max}}/T_{n,2})} < \frac{6400n^2 8}{2^{2n+3}} < 1,$$

shows that the first endpoint always dominates. Moreover

$$\frac{\lfloor (n+1)/2 \rfloor T_{n+1,2}}{\lfloor n/2 \rfloor T_{n,2}} \leq \frac{31}{29} \frac{11}{10} \left(\frac{31}{30}\right)^2 \frac{1}{4} < \frac{1}{3}. \quad (13)$$

Consequently the sum of all row perturbations after row 30 satisfies

$$S := \sum_{n \geq 30} \left(\frac{10}{2^n} + \sum_{k < n} |A_{n,k}| \right) < 1.883 \times 10^{-8}. \quad (14)$$

The forcing occurs only for odd n , and its two-step ratio is $100 2^{-(2n+3)}$; hence

$$H := \sum_{n \geq 30} |G_n| < 9.78 \times 10^{-118}. \quad (15)$$

All coefficients through order 30 are rational: the Taylor coefficients of $\sin$, $\arctan$, and (5) are rational when $\varepsilon = 1/10$. The included exact-arithmetic certificate computes

$$w_{30} = a_{30}y(0) + b_{30}, \quad a_{30} \doteq 109.81990128428, \quad b_{30} \doteq 349.05750538187. \quad (16)$$

By Lemma 1, $|y(0)| \leq 10/9$. Exact rational comparisons in the script prove

$$|w_{30}| > 227, \quad \max_{k \leq 30} |w_k| < 735. \quad (17)$$

If $W_N = \max_{k \leq N} |w_k|$, equations (8)–(15) imply that all later terms are bounded by $(W_{30} + H)/(1 - S)$ and

$$|w_\infty - w_{30}| \leq \frac{S(W_{30} + H)}{1 - S} + H < 10^{-3}. \quad (18)$$

Thus w_n converges and $|w_\infty| > 226.999$ for the unique periodic solution at $\varepsilon = 1/10$.

If this solution were analytic at zero, its Taylor coefficients would have bounded n -th roots. Instead, (7) and $w_n \rightarrow w_\infty \neq 0$ give

$$|y_n|^{1/n} = |\varepsilon| \frac{2^{(n-1)/2}}{(n!)^{1/n}} |w_n|^{1/n} \rightarrow \infty,$$

a contradiction. Hence $y_{1/10}$ is C^∞ but nonanalytic.

Why the parameter can be arbitrarily small

The source's persistence radius need not contain $1/10$, so one final point is essential. On the complex Banach space of continuous periodic functions,

$$y_\varepsilon = (I - \varepsilon A^{-1}C)^{-1} A^{-1} \sin$$

is holomorphic for $|\varepsilon| < 1$. On every compact subset of the punctured unit disc, the estimates above (with $|\varepsilon|$ bounded away from zero) make the sequence $w_n(\varepsilon)$ locally uniformly convergent. Therefore $W(\varepsilon) := \lim_n w_n(\varepsilon)$ is holomorphic on $0 < |\varepsilon| < 1$. Since the exact certificate proves $W(1/10) \neq 0$, the identity theorem says that W cannot vanish on any interval $(0, \delta)$. There are consequently positive parameters $\varepsilon \rightarrow 0$ for which $W(\varepsilon) \neq 0$ and y_ε is nonanalytic.

Choose such an ε below the source theorem's persistence threshold. The clock subsystem has only the unit-circle periodic orbit nearby, up to phase, and Lemma 1 makes its z component unique. Hence $K_\varepsilon = (\cos t, \sin t, y_\varepsilon(t))$ is exactly the locally persisted branch (after the source's phase normalization). It is C^∞ , by the bootstrap, but not analytic. We have proved the following.

Theorem 2. *There are real-analytic f, P, r satisfying the state-dependent-delay setup of arXiv:2103.05203 and arbitrarily small positive perturbation parameters ε for which the periodic orbit persisted from a nondegenerate analytic ODE orbit is C^∞ but not real analytic. Thus the natural question in Remark 6.5 has a negative answer, and the authors' conjectured failure of analyticity occurs already in dimension three with one bounded positive delay.*

Verification and literature boundary

Run the exact certificate from this directory with

```
conda run --no-capture-output -n sandbox python code/rigorous_certificate.py
```

It uses `Fractions.Fraction`; floating-point numbers are used only for display after every decisive inequality has been checked exactly.

Mallet-Paret and Nussbaum established analogous expansive-fixed-point obstructions for analytic equations with prescribed time-dependent delays [2], and their paper explicitly separates the state-dependent case. Hu proves analyticity for a special state-dependent system under a monotone time-map hypothesis [3]; that hypothesis excludes the expanding fixed point used here. Searches through 12 August 2026 found no paper resolving Remark 6.5 itself. This is a bounded novelty check, not a claim of exhaustive bibliographic priority.

References

- [1] J. Yang, J. Gimeno, and R. de la Llave, *Persistence and smooth dependence on parameters of periodic orbits in functional differential equations close to an ODE or an evolutionary PDE*, arXiv:2103.05203 (2021), Remark 6.5.
- [2] J. Mallet-Paret and R. D. Nussbaum, *Analyticity and nonanalyticity of solutions of delay-differential equations*, SIAM J. Math. Anal. 46 (2014), arXiv:1305.0579.
- [3] Z. Hu, *Analyticity of bounded solutions of analytic state-dependent delay differential equations*, arXiv:1708.08024 (2017).